

1. `sudo apt update`
2. `sudo apt install python3-pip -y`
3. `pip3 install edge-impulse-linux flash --break-system-packages`
4. `sudo apt update`
5. `sudo apt install libasound-dev libportaudio2 libportaudiocpp0 portaudio19-dev -y`
6. `pip install pyaudio --break-system-packages`
7. `pip install six --break-system-packages`
8. `sudo apt update`
9. `sudo apt install libgl1 libglvnd0 libglx-mesa0 -y`
10. `pip3 install "opencv-python>=4.5.1.48,<5" --break-system-packages`

Step 2

Create the model.eim

Transfer model.eim to edgserver

```
scp model.eim raspberry@192.168.10.119:/home/raspberry/model.eim
```

Step 3

edgserver script

```
from flask import Flask, request, jsonify
from edge_impulse_linux.runner import ImpulseRunner
import os
```

```
app = Flask(__name__)
```

```
# Use the absolute path we verified yesterday
```

```
model_file = '/home/raspberry/model.eim'
```

```
# Initialize the runner
```

```
runner = ImpulseRunner(model_file)
runner.init()
```

```
@app.route('/predict', methods=['POST'])
def predict():
```

```

try:
    data = request.json
    features = data['features']

    # Run the AI model
    res = runner.classify(features)

    # 'result' contains the scores for IDLE and FAN_ON
    predictions = res['result']['classification']

    # Pick the label with the highest score
    label = max(predictions, key=predictions.get)
    confidence = predictions[label]

    print(f"Input: {features} -> Prediction: {label}
    ({confidence*100:.2f}%)")

    return jsonify({
        "label": label,
        "confidence": predictions # Returns scores for both classes
    })
except Exception as e:
    return jsonify({"error": str(e)}), 400

if __name__ == '__main__':
    # Listen on all network interfaces on port 5000
    app.run(host='0.0.0.0', port=5000)

```

Note: while running the above script you may encounter this error

Traceback (most recent call last):

```

File "/home/raspberry/edgeserver.py", line 12, in <module>
    runner.init()
File
"/home/raspberry/.local/lib/python3.13/site-packages/edge_impulse_linux/run
ner.py", line 34, in init
    raise Exception('Model file "' + self._model_path + '" is not
executable')
Exception: Model file "/home/raspberry/model.eim" is not executable

```

This error means that even though the file is there, Linux is blocking it from running for safety. You need to give it execution permissions. Think of it like a new app—Ubuntu requires you to "unlock" it before it can

actually perform tasks.

Run this single command in your Raspberry Pi terminal:

```
bash
chmod +x /home/raspberry/model.eim
```

Running the chmod script and then the script again produce this result in the terminal

```
* Serving Flask app 'edgeserver'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production
deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.10.119:5000
Press CTRL+C to quit
```

Edge Server is now active and ready to work with Pico W sending sensor data to it